Enhanced RSS-Based UAV Localization Via Trajectory and Multi-Base Stations

Yifan Li, Feng Shu, *Member, IEEE*, Baihua Shi, Xin Cheng,
Yaoliang Song, and Jiangzhou Wang, *Fellow, IEEE*

***Abstract*— To improve the localization precision of unmanned aerial vehicle (UAV), a novel framework is established by jointly utilizing multiple measurements of received signal strength (RSS) from multiple base stations (BSs) and multiple points on trajectory. First, a joint maximum likelihood (ML) of exploiting both trajectory information and multi-BSs is proposed. To reduce its high complexity, two low-complexity localization methods are designed. The first method is from BS to trajectory (BST), called LCSL-BST. First, fixing the nth BS, by exploiting multiple measurements along trajectory, the position of UAV is computed by ML rule. Finally, all computed positions of UAV for different BSs are combined to form the resulting position. The second method reverses the order, called LCSL-TBS. We also derive the Cramer-Rao lower boundary (CRLB) of the joint ML method. From simulation results, we can see that the proposed joint ML and separate LCSL-BST methods have made a significant improvement over conventional ML method without use of trajectory knowledge in terms of location performance. The former achieves the joint CRLB and the latter is of low-complexity.**

***Index Terms*— Unmanned aerial vehicle (UAV), received signal strength (RSS), trajectory localization, maximum likelihood (ML), Cramer-Rao lower boundary (CRLB).**

I. Introduction

IN RECENT years, unmanned aerial vehicle (UAV) has been a hot topic, and technologies related to it are rapidly developed. Since the advantages in terms of three-dimensional (3D) mobility and flexible deployment, UAVs have been used in civilian and military areas, such as taking aerial photographs or videos, detecting enemy situations and destroying specific targets [1]. Although currently there are not many actual cases of large-scale use of UAVs in the wireless communications, according to existing studies, the problem of physical layer security in UAV networks has attracted great attention [2], [3]. Thus, when UAVs appear above our city, a critical question we need to consider is how to locate them quickly and precisely. The position information of UAV is very helpful for us to plan the flight trajectory of UAV, so that we can cover a larger communication range with fewer UAVs.

Manuscript received February 3, 2021; accepted February 16, 2021. Date of publication February 22, 2021; date of current version June 10, 2021. This work was supported in part by the National Natural Science Foundation of China (Nos. 62071234 and 61771244). The associate editor coordinating the review of this letter and approving it for publication was S. Bartoletti. *(Corresponding authors: Feng Shu; Yaoliang Song.)*

Yifan Li, Baihua Shi, Xin Cheng, and Yaoliang Song are with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: ylsong@mail.njust.edu.cn).

Feng Shu is with the School of Information and Communication Engineering, Hainan University, Haikou 570228, China, and also with the School of Electronic and Optical Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: shufeng0101@163.com).

Jiangzhou Wang is with the School of Engineering and Digital Arts, University of Kent, Canterbury CT2 7NT, U.K. (e-mail: j.z.wang@kent.ac.uk).

Digital Object Identifier 10.1109/LCOMM.2021.3061104

In order to obtain the position information of UAV accurately, the appropriate localization technology should be selected. Over the past few decades, global positioning system (GPS) has been the preferred technology in positioning or navigation because of its reliability and accuracy in outdoor open situations. While in urban environments where buildings and people are crowded, or in battlefields, GPS signals are often weak or even blocked [4]. In these situations, we should consider the use of signals transmitted by the base stations (BSs) to determine the position of target. So we have turned our attention to other more practical geolocation techniques, including received signal strength (RSS), angle of arrival (AOA), time of arrival (TOA), and time difference of arrival (TDOA) [5]–[7]. This letter mainly focuses on RSS, because of the following advantages: only a single antenna being required, simpler receivers, lower in cost, and low power consumption [8].

In order to ensure that the UAV can obtain its own location information as accurately as possible, we propose a 3D UAV self-positioning system model which is different from the traditional RSS-based 2D positioning model [9]. In this model, a moving UAV combines trajectory information and RSS measurements from surrounding BSs to locate itself. Our main contributions are summarized as follows:

1) The traditional model [9] uses the RSS measurements from the surrounding base stations to locate a fixed target in a 2D plane, so the gain of the system only comes from the base station dimension. While in our model, the UAV flies along a 3D trajectory and receives RSS measurements from multiple base stations at each time during the flight. Therefore, compared with traditional model, we take advantage of the UAV's flexibility so that the system obtains additional time and space gain.
2) To improve the localization accuracy, with the help of trajectory knowledge, a joint UAV ML localization method is proposed in multi-BS scenario. Compared to the case without use of trajectory knowledge, the proposed joint method makes a significant improvement in positioning accuracy. Then, the corresponding Cramer Rao lower bound (CRLB) is derived. Simulation results show that, the proposed method can achieve the joint CRLB.
3) To reduce the high computational complexity of the proposed joint localization method above, two low-complexity separate localization (LCSL) methods are proposed. The first method is from BS to trajectory (BST), called LCSL-BST. First, fixing the nth BS, by exploiting multiple measurements along trajectory, the position of UAV is computed by ML rule. Finally,

1558-2558 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:45:59 UTC from IEEE Xplore. Restrictions apply.

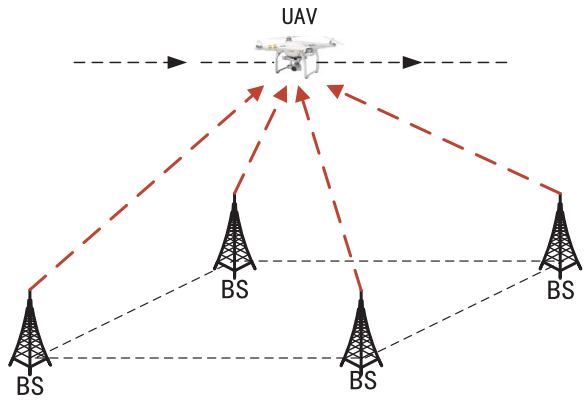


Fig. 1. The UAV moves along the trajectory and receives signals from surrounding base stations.

all computed positions of UAV for different BSs are combined to form the resulting position. Similarly, the second method is from trajectory to BS (TBS), called LCSL-TBS. Compared with the joint UAV ML localization, the computational complexities of the two methods are reduced significantly. For large values of measurement standard deviation σ, the LCSL-BST makes a significant improvement over the ML without using trajectory knowledge.

Notation: Matrices, vectors, and scalars are denoted by letters of bold upper case, bold lower case, and lower case, respectively. Signs $(\cdot)^T$, $|\cdot|$ and $\|\cdot\|$ represent transpose, modulus and norm. $\mathbf{J}_{l\times m}$ represents a $l \times m$ matrix with all elements equal to 1 and $K \times K$ identity matrix is denoted by $\mathbf{I}_K$. The expectation operator is denoted by $E[\cdot]$. $\otimes$ denotes the Kronecker product, $\mathrm{vec}(\cdot)$ denotes vec operator, $\mathrm{tr}(\cdot)$ represents the trace of matrix and diagonal matrix is denoted by $\mathrm{diag}(\cdot)$.

II. System Model

As is presented in Fig.1, an unmanned aerial vehicle (UAV) flies along an arbitrary trajectory from an unknown initial position and receives signals from the surrounding N BSs during the flight. Assuming that the BSs are located at $\mathbf{s}_n = [x_n^*, y_n^*, z_n^*], n = 1, 2, \cdots, N$, and the position of the UAV at time k is $\mathbf{u}_k = [x_k, y_k, z_k], k = 1, 2, \cdots, K$. Supposing the time interval between any two adjacent points on the UAV trajectory is Δt_i, $i = 1, 2, \cdots, K-1$, and the UAV flies at a velocity of $\mathbf{v}_i = [v_{x,i}, v_{y,i}, v_{z,i}]$ at Δt_i. The velocity information contains direction and speed, which can be obtained by the compass and speed sensors carrying on the UAV. Then position of the UAV at time k can be represented by the position and velocity information of previous time as

$$\mathbf{u}_k = \mathbf{u}_{k-1} + \mathbf{v}_{k-1}\Delta t_{k-1} = \mathbf{u}_1 + \sum_{i=1}^{k-1} \mathbf{v}_i \Delta t_i \tag{1}$$

Since only at time K the UAV can get the complete velocity information from time 1 to time K, the UAV's position can also be related to $\mathbf{u}_K$ as follows

$$\mathbf{u}_k = \mathbf{u}_K - \sum_{i=k}^{K-1} \mathbf{v}_i \Delta t_i. \tag{2}$$

Using the log-distance path loss propagation model [10], [11], the RSS measurements coming from the nth BS and received by the UAV at time k, is given by

$$r_{k,n} = \alpha_n + a_{k,n} + w_{k,n}, \tag{3a}$$

$$a_{k,n} = -10\gamma \log_{10} d_{k,n}, \tag{3b}$$

where α_n denotes the transmitted signal power of the nth BS, and is assumed to be unknown. γ stands for the path loss exponent, and its values generally range from 2 to 5 [12]. $w_{k,n}$ is a zero-mean Gaussian random variable with known standard deviation $\sigma_{k,n} > 0$. The Euclidean distance between the nth BS and the UAV at time k can be verified as

$$\begin{aligned} d_{k,n} &= \|\mathbf{u}_k - \mathbf{s}_n\| \\ &= \|\mathbf{u}_K - (\mathbf{s}_n + \Delta\mathbf{u}_k)\| \\ &= \|\mathbf{u}_K - \mathbf{s}_{k,n}\|. \end{aligned} \tag{4}$$

where $\Delta\mathbf{u}_k = \sum_{i=k}^{K-1} \mathbf{v}_i \Delta t_i$. From another perspective, the system model can be viewed as the UAV fixed at $\mathbf{u}_k$, while the nth BS moves from the initial position $\mathbf{s}_n$ to the $\mathbf{s}_{k,n}$ at time k. Obviously, the BS doesn't move in reality, so they are called virtual moving BSs in our assumption.

Collecting all measurement values from N virtual moving BSs at all K times forms the following $K \times N$ RSS matrix

$$\mathbf{R} = \boldsymbol{\alpha} \otimes \mathbf{J}_{K\times 1} + \mathbf{A} + \mathbf{W}, \tag{5}$$

where $\mathbf{R} = [r_{k,n}]_{K\times N}$, $\mathbf{A} = [a_{k,n}]_{K\times N}$, $\mathbf{W} = [w_{k,n}]_{K\times N}$ and $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \cdots, \alpha_N]$.

In order to obtain the maximum likelihood (ML) estimator of parameters, using the vec operator to transform the matrices to vectors yields the following form

$$\mathbf{r} = \boldsymbol{\alpha}^T \otimes \mathbf{J}_{K\times 1} + \mathbf{a} + \mathbf{w}, \tag{6}$$

where $\mathbf{r} = \mathrm{vec}(\mathbf{R})$, $\mathbf{a} = \mathrm{vec}(\mathbf{A})$ and $\mathbf{w} = \mathrm{vec}(\mathbf{W})$. Based on the basic theorems of the variance,

$$E(w_{k_1,n_1} w_{k_2,n_2}) = \begin{cases} \sigma^2_{k_1,n_1}, & k_1 = k_2 \quad and \quad n_1 = n_2 \\ 0, & k_1 \neq k_2 \quad or \quad n_1 \neq n_2 \end{cases} \tag{7}$$

where $1 \le k_1, k_2 \le K$ and $1 \le n_1, n_2 \le N$, the covariance matrix of $\mathbf{w}$ is given by

$$\begin{aligned} \mathbf{C} &= E[(\mathbf{w} - E[\mathbf{w}])(\mathbf{w} - E[\mathbf{w}])^T] \\ &= \mathrm{diag}\{\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_K\}, \end{aligned} \tag{8}$$

where $\mathbf{V}_k = \mathrm{diag}\{\sigma^2_{k,1}, \sigma^2_{k,2}, \cdots, \sigma^2_{k,N}\}$.

III. Proposed Localization Methods With the Aid of Trajectory

In this section, to achieve a high-precision localization, three methods are proposed: an optimal joint ML method and two low-complexity separate methods. The latter two methods make a position estimate along BS dimension and the trajectory dimension separately to achieve a low-complexity at the expense of some performance loss.

A. Proposed Joint Trajectory-Aided ML Localization Method

According to the system model, the localization of the UAV position $\mathbf{u}_K$ is based on RSS measurements from N BSs at all K times and assuming that the reference power of each BS

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:45:59 UTC from IEEE Xplore. Restrictions apply.

is equal to α. The probability density function (pdf) of RSS measurements $\mathbf{r}$ given $\mathbf{u}_K$ and α, is denoted by

$$\begin{aligned} f(\mathbf{r}|\mathbf{u}_K,\alpha) &= M\exp\Bigg\{-\frac{1}{2}(\mathbf{r}-\mathbf{a}-\alpha\mathbf{J}_{NK\times 1})^T\mathbf{C}^{-1} \\ &\quad\cdot(\mathbf{r}-\mathbf{a}-\alpha\mathbf{J}_{NK\times 1})\Bigg\} \\ &= M\exp\Bigg\{-\frac{1}{2}\sum_{k=1}^{K}\sum_{n=1}^{N}\sigma_{k,n}^{-2}(r_{k,n}-a_{k,n}-\alpha)^2\Bigg\}, \quad (9) \end{aligned}$$

where $M = \left((2\pi)^{NK}|\mathbf{C}|\right)^{-\frac{1}{2}}$. Given measurement data $\mathbf{r}$, the above function is the likelihood function of two parameters $\mathbf{u}_K$ and α. Maximizing the likelihood function with respect to $\mathbf{u}_K$ and α is equivalent to minimizing the function

$$Q(\mathbf{u}_K,\alpha)=\sum_{k=1}^{K}\sum_{n=1}^{N}\sigma_{k,n}^{-2}(r_{k,n}-a_{k,n}-\alpha)^2. \quad (10)$$

Fixing parameter $\mathbf{u}_K$, the optimal value $\hat{\alpha}$ of minimizing the above function is given by

$$\hat{\alpha}=\frac{1}{NK}\sum_{k=1}^{K}\sum_{n=1}^{N}(r_{k,n}-a_{k,n}). \quad (11)$$

Substituting (11) back in (10) yields

$$Q(\mathbf{u}_K)=\sum_{k=1}^{K}\sum_{n=1}^{N}\sigma_{k,n}^{-2}\Bigg(r_{k,n}-a_{k,n} -\frac{1}{NK}\sum_{k=1}^{K}\sum_{n=1}^{N}(r_{k,n}-a_{k,n})\Bigg)^2. \quad (12)$$

Maximizing $Q(\mathbf{u}_K)$ to obtain the optimal $\hat{\mathbf{u}}_K$ is done by performing the following nonlinear optimization

$$\hat{\mathbf{u}}_K=\operatorname{argmin}\left\{Q(\mathbf{u}_K)\right\}, \quad (13)$$

where $\hat{\mathbf{u}}_K = [\hat{x}_K,\hat{y}_K,\hat{z}_K]^T$. For this nonlinear optimization problem, the grid search is a reliable solution. The computational complexity of each search is $4(KN)^3$ real multiplications.

B. Proposed LCSL-BST

Suppose the UAV only receives signals from one BS at all K times. So if we want to locate the UAV's position according to all N BSs, each BS is virtually viewed as moving along the trajectory separately. Finally, the result is a weighted combination of these N estimations. Assuming the RSS measurement $\mathbf{r}_n$ from the nth BS is

$$\mathbf{r}_n=\alpha\mathbf{J}_{K\times 1}+\mathbf{a}_n+\mathbf{w}_n, \quad (14)$$

where $\mathbf{r}_n=[r_{1,n},\cdots,r_{K,n}]^T$, $\mathbf{a}_n=[a_{1,n},\cdots,a_{K,n}]^T$, and $\mathbf{w}_n=[w_{1,n},\cdots,w_{K,n}]^T$. The objective function (12) is rewritten as

$$Q_n(\mathbf{u}_K)=\sum_{k=1}^{K}\sigma_{k,n}^{-2}\Bigg(r_{k,n}-a_{k,n}-\frac{1}{K}\sum_{k=1}^{K}(r_{k,n}-a_{k,n})\Bigg)^2, \quad (15)$$

and then perform the optimization based on the RSS measurement from nth BS at all K times

$$\tilde{\mathbf{u}}_{K,n}=\operatorname{argmin}\left\{Q_n(\mathbf{u}_1)\right\}. \quad (16)$$

Since each estimate is based on RSS measurements from different BSs, the location result after the nth estimate, corresponding to the nth BS, is designed as the combination of the current estimate $\tilde{\mathbf{u}}_{K,n}$ and the location result $\hat{\mathbf{u}}_{K,n-1}$ after the previous estimate. Then we can get $\hat{\mathbf{u}}_{K,n}=f_n(\hat{\mathbf{u}}_{K,n-1},\tilde{\mathbf{u}}_{K,n})$, where function $f_n()$ denotes a linear combination and the weighting coefficients are assumed to be equal for each term. So, the estimation result associated with the nth BS is

$$\begin{aligned} f_n(\hat{\mathbf{u}}_{K,n-1},\tilde{\mathbf{u}}_{K,n}) &= \frac{1}{n}\sum_{m=1}^{n}\tilde{\mathbf{u}}_{K,m} \\ &= \frac{n-1}{n}\hat{\mathbf{u}}_{K,n-1}+\frac{1}{n}\tilde{\mathbf{u}}_{K,n}, \quad (17) \end{aligned}$$

where $\hat{\mathbf{u}}_{K,n-1}=\frac{1}{n-1}\sum_{m=1}^{n-1}\tilde{\mathbf{u}}_{K,m}$ and the final positioning result is

$$\hat{\mathbf{u}}_K=f_N(\hat{\mathbf{u}}_{K,N-1},\tilde{\mathbf{u}}_{K,N})=\frac{N-1}{N}\hat{\mathbf{u}}_{K,N-1}+\frac{1}{N}\tilde{\mathbf{u}}_{K,N}, \quad (18)$$

where $\hat{\mathbf{u}}_K=[\hat{x}_K,\hat{y}_K,\hat{z}_K]^T$. The grid search steps for this method are similar to the previous one, with a complexity of $4K^3N$ real multiplications per search.

C. Proposed LCSL-TBS

After getting the trajectory information at time K, we assume that the UAV locates itself separately according to the RSS measurements from the virtual BSs at each time. The RSS measurements obtained at time k is

$$\mathbf{r}_k=\alpha\mathbf{J}_{N\times 1}+\mathbf{a}_k+\mathbf{w}_k, \quad (19)$$

where $\mathbf{r}_k=[r_{k,1},\cdots,r_{k,N}]^T$, $\mathbf{a}_k=[a_{k,1},\cdots,a_{k,N}]^T$ and $\mathbf{w}_k=[w_{k,1},\cdots,w_{k,N}]^T$. The object function Q at time k is given by

$$Q_k(\mathbf{u}_K)=\sum_{n=1}^{N}\sigma_{k,n}^{-2}\Bigg(r_{k,n}-a_{k,n}-\frac{1}{N}\sum_{n=1}^{N}(r_{k,n}-a_{k,n})\Bigg)^2, \quad (20)$$

then, the optimization performed at time k is

$$\tilde{\mathbf{u}}_{K,k}=\operatorname{argmin}\left\{Q_k(\mathbf{u}_K)\right\}. \quad (21)$$

Similar to the previous section, combining $\tilde{\mathbf{u}}_{K,k}$ at all the K time generates the final result

$$\hat{\mathbf{u}}_K=\frac{1}{K}\sum_{k=1}^{K}\tilde{\mathbf{u}}_{K,k}=\frac{K-1}{K}\hat{\mathbf{u}}_{K,K-1}+\frac{1}{K}\tilde{\mathbf{u}}_{K,K}, \quad (22)$$

where $\hat{\mathbf{u}}_{K,K-1}=\frac{1}{K-1}\sum_{m=1}^{K-1}\tilde{\mathbf{u}}_{K,m}$. The computational complexity of each search is $4KN^3$ real multiplications.

IV. Simulation Results and Discussion

To illustrate the concepts and algorithms discussed in this letter, we will show some numerical results of the proposed methods in this section. According to the system model and equation (13), grid search is a reliable method to find the point $\hat{\mathbf{u}}_1$ that minimizes the objective function Q. Obviously, the complexity of 3D grid search is very high. In order to facilitate the simulation, the UAV is set to fly at a known altitude, thus turning the problem into a 2D grid search. In the

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:45:59 UTC from IEEE Xplore. Restrictions apply.

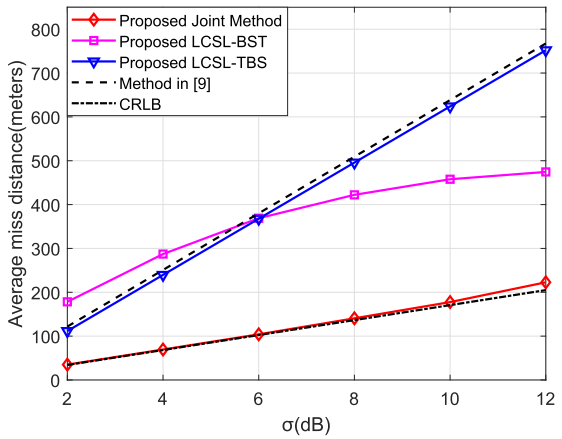


Fig. 2. Average miss distance versus the measurement standard deviation σ.

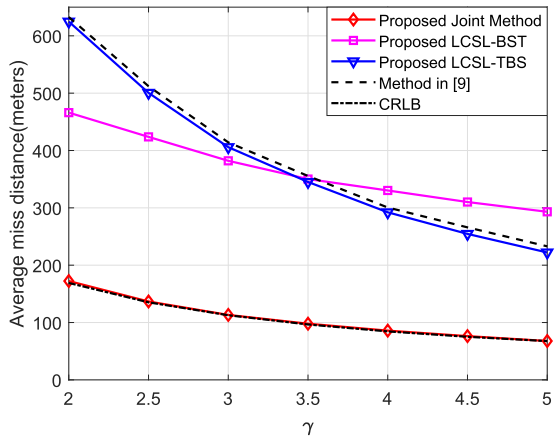


Fig. 3. Average miss distance versus the path loss exponent γ.

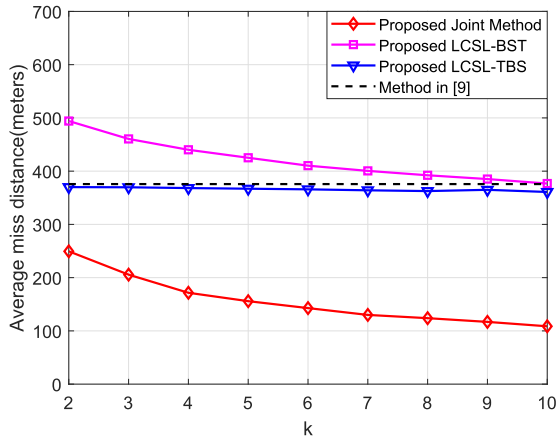


Fig. 4. Positioning accuracy of the proposed methods at each time during the flight.

simulations, the UAV's trajectory is generated randomly, $\mathbf{v}_i \leq$ 5m/s, flight altitude is $z = 50$m, $K = 10$, and $\Delta t_1 = \cdots = \Delta t_{K-1} = 5$s. Since the flying velocity is slow and sampling time interval is short, the position change caused by the velocity change of the UAV is generally smaller than positioning error. So the change of the UAV's velocity will not have much impact on the actual positioning result. For the BSs, $N = 6$ and BSs are located on the corners of a hexagon. The distance between two adjacent BSs is 1km and the height of each BS is 20 meters. The error terms $w_{k,n}$ in (3a) are all simulated as independent zero-mean Gaussian random variables with identical variance σ^2. To determine the $\hat{\mathbf{u}}_K$, a 2km×2km area of interest (AOI) is set up and the area is divided into a 10m×10m grid. The miss distance is the distance between the $\hat{\mathbf{u}}_K$ and $\mathbf{u}_K$. The average miss distance is computed from 1000 miss distance simulation runs.

Fig.2 plots the average miss distance as a function of the measurement standard deviation σ, where $\gamma = 3.3$. The proposed joint method makes a significant improvement in the positioning accuracy over existing method [9]. More importantly, the proposed joint method can achieve the joint CRLB. The proposed LCSL-TBS is slightly better than existing method [9] for almost all values of measurement standard deviation σ. The proposed LCSL-BST performs better than existing method [9] for large values of measurement standard deviation σ, for example, σ is larger than 6dB. The running time required for each search of the joint method, LCSL-BST, LCSL-TBS and method in [9] is 2.75s, 0.55s, 0.44s and 0.41s.

Fig.3 demonstrates how the localization accuracy increases with the increase of the path loss exponent γ, where $\sigma = 6$dB. The trend of the curve seems to be reversed because a larger path loss factor weakens the received signal strength, while this can be interpreted as that with the increase of γ, the signal strength will be more sensitive to the change of distance. In particular, the proposed LCSL-BST performs better than existing method [9] for small values of path loss exponent γ.

Fig.4 gives the positioning accuracy of the proposed methods at each time from time 2 to time K, where $\sigma = 6$dB and $\gamma = 3.3$. The positioning accuracy of joint method at each time is much better than method in [9], and it is also gradually improved with the increase in time. The proposed LCSL-TBS is slightly better than existing method in [9] at each time. Since LCSL-BST has a poor performance when $\sigma \leq 6$dB, as shown in Fig.2, the performance advantages of this method cannot be reflected under such parameter condition.

V. Conclusion

In our work, an enhanced RSS-based UAV localization model has been proposed. This model jointly exploits the trajectory information and RSS measurements from multiple BSs to improve the UAV localization performance. First, a joint ML localization method and two low-complexity separate methods were proposed in section III. The joint CRLB was also derived as a performance benchmark. The simulation results have shown that, compared with existing method without use of trajectory knowledge, the proposed joint method makes a significant improvement in positioning accuracy. The LCSL-BST and LCSL-TBS achieve a low-complexity at the expense of performance loss compared to the joint method.

Appendix: Derivation of Joint CRLB

In order to obtain the Cramer-Rao lower bound (CRLB) on the location-error variance of the joint ML trajectory method, we extend the derivation in [9] to our work. Assuming $\mathbf{C} = \sigma^2 \mathbf{I}_{NK}$, then

$$\text{CRLB} = \sigma^2 \mathbf{G}^{-1}, \tag{23}$$

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:45:59 UTC from IEEE Xplore. Restrictions apply.

where $\mathbf{G}$ is a 3×3 symmetrical matrix and its elements can be expressed as

$$\begin{aligned} g_{i,j} &= \mathbf{a}_{\mathbf{u}_K(i)}^T \mathbf{a}_{\mathbf{u}_K(j)} - \frac{1}{NK} \sum \mathbf{a}_{\mathbf{u}_K(i)} \sum \mathbf{a}_{\mathbf{u}_K(j)} \\ &= p_{i,j} - q_{i,j}, \end{aligned} \tag{24}$$

where $\mathbf{u}_K = [x_K, y_K, z_K]$, $1 \le i, j \le 3$, $\mathbf{u}_K(i)$ or $\mathbf{u}_K(j)$ represents the ith or jth element of the vector $\mathbf{u}_K$. Then equations (3a) and (4) lead to the following $NK \times 1$ vectors

$$\mathbf{a}_{\mathbf{u}_K(i)} = \frac{\partial \mathbf{a}}{\partial \mathbf{u}_K(i)} = \beta[\mathbf{a}_{\mathbf{u}_K(i),1}, \mathbf{a}_{\mathbf{u}_K(i),2}, \ldots, \mathbf{a}_{\mathbf{u}_K(i),K}]^T \tag{25}$$

where $\beta = -(10\gamma)/\ln 10$. Let $\Delta\mathbf{u}_k = [\Delta x_k, \Delta y_k, \Delta z_k]$, we can get

$$\mathbf{a}_{\mathbf{u}_K(1),k} = \left[\frac{x_K - \Delta x_k - x_1^*}{d_{k,1}^2}, \ldots, \frac{x_K - \Delta x_k - x_N^*}{d_{k,N}^2} \right], \tag{26a}$$

$$\mathbf{a}_{\mathbf{u}_K(2),k} = \left[\frac{y_K - \Delta y_k - y_1^*}{d_{k,1}^2}, \ldots, \frac{y_K - \Delta y_k - y_N^*}{d_{k,N}^2} \right], \tag{26b}$$

$$\mathbf{a}_{\mathbf{u}_K(3),k} = \left[\frac{z_K - \Delta z_k - z_1^*}{d_{k,1}^2}, \ldots, \frac{z_K - \Delta z_k - z_N^*}{d_{k,N}^2} \right]. \tag{26c}$$

Then we can get

$$\begin{cases} p_{1,1} = \beta^2 \sum_{k=1}^K \sum_{n=1}^N \dfrac{(x_K - \Delta x_k - x_n^*)^2}{d_{k,n}^4} \\ p_{1,2} = \beta^2 \sum_{k=1}^K \sum_{n=1}^N \dfrac{(x_K - \Delta x_k - x_n^*)(y_K - \Delta y_k - y_n^*)}{d_{k,n}^4} \\ \quad \vdots \\ p_{3,3} = \beta^2 \sum_{k=1}^K \sum_{n=1}^N \dfrac{(z_K - \Delta z_k - z_n^*)^2}{d_{k,n}^4} \end{cases} \tag{27}$$

and

$$\begin{cases} q_{1,1} = \dfrac{\beta^2}{NK} \left(\sum_{k=1}^K \sum_{n=1}^N \dfrac{x_K - \Delta x_k - x_n^*}{d_{k,n}^2} \right)^2 \\ q_{1,2} = \dfrac{\beta^2}{NK} \sum_{k=1}^K \sum_{n=1}^N \dfrac{x_K - \Delta x_k - x_n^*}{d_{k,n}^2} \\ \qquad \times \sum_{k=1}^K \sum_{n=1}^N \dfrac{y_K - \Delta y_k - y_n^*}{d_{k,n}^2} \\ \quad \vdots \\ q_{3,3} = \dfrac{\beta^2}{NK} \left(\sum_{k=1}^K \sum_{n=1}^N \dfrac{z_K - \Delta z_k - z_n^*}{d_{k,n}^2} \right)^2 \end{cases} \tag{28}$$

The average miss distance between the estimated position of UAV and the actual position is defined as

$$\delta = \sqrt{E[(\hat{x}_K - x_K)^2 + (\hat{y}_K - y_K)^2 + (\hat{z}_K - z_K)^2]}. \tag{29}$$

Since CRLB is the variance lower bound of unbiased estimator, let $\hat{\mathbf{u}}_K$ be an unbiased estimator of $\mathbf{u}_K$, i.e., $E(\hat{\mathbf{u}}_K) = \mathbf{u}_K$. According to the definition of covariance, the average miss distance satisfies the following inequality

$$\delta \ge \sqrt{\mathrm{tr}(\mathrm{CRLB})} = \sigma \sqrt{\mathrm{tr}(\mathbf{G}^{-1})} \tag{30}$$

where

$$\mathrm{tr}(\mathbf{G}^{-1}) = \frac{\sum_{i,j=1;i\neq j}^3 (g_{i,i} g_{j,j} - g_{i,j}^2)}{h} \tag{31a}$$

$$\begin{aligned} h = {} & g_{1,1}(g_{2,2}g_{3,3} - g_{2,3}^2) - g_{1,2}(g_{1,2}g_{3,3} - g_{1,3}g_{2,3}) \\ & + g_{1,3}(g_{1,2}g_{2,3} - g_{1,3}g_{2,2}). \end{aligned} \tag{31b}$$

The right hand of (30) is the CRLB for the average miss distance between $\hat{\mathbf{u}}_K$ and $\mathbf{u}_K$.

REFERENCES

[1] Y. Zeng, R. Zhang, and T. J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," *IEEE Commun. Mag.*, vol. 54, no. 5, pp. 36–42, May 2016.

[2] W. Wang *et al.*, "Joint precoding optimization for secure SWIPT in UAV-aided NOMA networks," *IEEE Trans. Commun.*, vol. 68, no. 8, pp. 5028–5040, Aug. 2020.

[3] N. Zhao *et al.*, "Security enhancement for NOMA-UAV networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 3994–4005, Apr. 2020.

[4] P. Wan, Q. Huang, G. Lu, J. Wang, Q. Yan, and Y. Chen, "Passive localization of signal source based on UAVs in complex environment," *China Commun.*, vol. 17, no. 2, pp. 107–116, Feb. 2020.

[5] F. Shu, S. Yang, Y. Qin, and J. Li, "Approximate analytic quadratic-optimization solution for TDOA-based passive multi-satellite localization with Earth constraint," *IEEE Access*, vol. 4, pp. 9283–9292, 2016.

[6] F. Shu, S. Yang, J. Lu, and J. Li, "On impact of Earth constraint on TDOA-based localization performance in passive multisatellite localization systems," *IEEE Syst. J.*, vol. 12, no. 4, pp. 3861–3864, Dec. 2018.

[7] F. Shu *et al.*, "Low-complexity and high-resolution DOA estimation for hybrid analog and digital massive MIMO receive array," *IEEE Trans. Commun.*, vol. 66, no. 6, pp. 2487–2501, Jun. 2018.

[8] S. Wang, B. R. Jackson, S. Rajan, and F. Patenaude, "Received signal strength-based emitter geolocation using an iterative maximum likelihood approach," in *Proc. IEEE Mil. Commun. Conf. (MILCOM)*, Nov. 2013, pp. 68–72.

[9] A. J. Weiss, "On the accuracy of a cellular location system based on RSS measurements," *IEEE Trans. Veh. Technol.*, vol. 52, no. 6, pp. 1508–1518, Nov. 2003.

[10] R. K. Martin, A. S. King, J. R. Pennington, R. W. Thomas, R. Lenahan, and C. Lawyer, "Modeling and mitigating noise and nuisance parameters in received signal strength positioning," *IEEE Trans. Signal Process.*, vol. 60, no. 10, pp. 5451–5463, Oct. 2012.

[11] D. Jin, F. Yin, C. Fritsche, F. Gustafsson, and A. M. Zoubir, "Bayesian cooperative localization using received signal strength with unknown path loss exponent: Message passing approaches," *IEEE Trans. Signal Process.*, vol. 68, pp. 1120–1135, 2020.

[12] B. R. Jackson, S. Wang, and R. Inkol, "Received signal strength difference emitter geolocation least squares algorithm comparison," in *Proc. 24th Can. Conf. Electr. Comput. Eng. (CCECE)*, May 2011, pp. 1113–1118.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on August 07,2026 at 08:45:59 UTC from IEEE Xplore. Restrictions apply.